

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

Fifth Semester of B. Tech. Examination (EC)

March/April 2018

EC 303.02/EC 303.01 Audio Video Engineering

Date: 31.03.2018, Saturday

Time: 01:30 p.m. To 04:30 p.m.

Maximum Marks: 70

Instructions:

1. The question paper comprises of two sections.
2. Section I and II must be attempted in separate answer sheets.
3. Make suitable assumptions and draw neat figures wherever required.
4. Use of scientific calculator is allowed.

<u>SECTION - I</u>		
Q 1	Do as Directed	[07]
(a)	Output of the carbon microphone is 100 mV with sound pressure of 1 microbar. In the absence of sound, the output reduces to 10 mV. Calculate signal to noise ratio.	[02]
(b)	Calculate the efficiency of loudspeaker if power intensity of sound is 0.08 W/m.m at 1 meter distance from the loud speaker, when the amplifier delivers 25 watt.	[02]
(c)	If normal conversation level is 25microW and we need at least 3dB more for comfortable listening in a meeting, calculate the intensity of sound required in dB over threshold of hearing ,presuming 3dB of power is absorbed by audience. Threshold of hearing is 1pW.	[03]
Q - 2 (a)	What are the essential requirements for good PA installation? How can these be met in practice?	[05]
(b)	What are the characteristics of a good loud speaker? Why efficiency of indirect radiating loudspeaker is higher than direct radiating type?	[04]
(c)	Describe Insulation to maintain intensity of sound.	[05]
<u>OR</u>		
Q - 2 (a)	Explain the Condenser type microphone.	[05]
(b)	What do you understand by woofer, squawker and tweeter speakers? Explain the necessity of cross over networks.	[04]
(c)	What are the requirements for a good auditorium for pleasant listening? Give salient features of acoustical design for an auditorium.	[05]
Q - 3	Write Short note on followings (Any Two)	[14]
(a)	3 D TV.	
(b)	Flat panel display TV Receivers.	
(c)	HDTV.	

<u>SECTION – II</u>		
Q - 4 (a)	Draw block diagram of a monochrome TV Receiver.	[04]
(b)	Justify the choice of a rectangular frame with width to height ratio equal to 4/3 for television transmission and reception.	[03]
Q - 5 (a)	What do you understand by interlaced scanning?	[05]
(b)	Sketch composite video signal waveform for at least three successive lines and indicate: (i) blanking level (ii) pedestal height (iii) sync pulse level.	[04]
(c)	Explain why FM is preferred for sound signal transmission. Describe how the use of pre-emphasis circuit at the transmitting end improves signal to noise ratio of the sound signal.	[05]
	<u>OR</u>	
Q - 5 (a)	Explain Vestigial Sideband Transmission.	[05]
(b)	Define Following. 1. Contrast 2. Brightness 3. HUE 4. Viewing Distance.	[04]
(c)	Write a short note on Interference suffered by TV signal.	[05]
Q - 6 (a)	Sketch the horizontal line and sync details. Also explain importance of Front Porch and Back Porch.	[06]
(b)	Justify the choice of 625 lines for TV transmission. Why is the total number of lines kept odd in all TV system?	[04]
(c)	Explain SAW Filter.	[04]
	<u>OR</u>	
Q – 6 (a)	Explain VHF Tuner .	[06]
(b)	Explain Adjacent channel interference.	[04]
(c)	Which are the merits of negative modulation?	[04]
